

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B. Tech. (EE) Examination

April-May 2013

EE205 Electrical Machine-I

Date: 23.04.2013, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Derive the expression of torque developed in doubly excited magnetic system. [07]
- Q - 2 (a) Explain in detail (i) Energy Balance (ii) Coupling field reaction [05]
- (b) Explain the working principle of DC generator. Also derive its EMF equation. [05]
- (c) Define the following terms : (i) Coil side (ii) Pole Pitch (iii) Front Pitch (iv) Back Pitch [04]

OR

- Q - 2 (a) Explain armature reaction with the help of suitable diagrams. [07]
- (b) An 8-pole DC shunt generator has 778 wave-connected armature conductors running at 500 rpm, supplies a load of 12.5Ω resistance at a terminal voltage of 250 V. The armature resistance is 0.24Ω and the field resistance is 250Ω . Find out the armature current, the induced EMF and the flux per pole. [05]
- (c) What do you mean by voltage regulation of a DC generator? [02]
- Q - 3 (a) Give the classification of DC motors with neat sketch and necessary equations. [06]
- (b) A 230 V DC motor takes no-load armature current of 2 A and runs at a speed of 1200 rpm. If the full-load current is 40 A find (i) the speed on full-load (ii) Percentage speed regulation. Resistance of armature is 0.25Ω . Assume that flux remains constant. [04]
- (c) What do you mean by torque? Explain armature torque and shaft torque. [04]

OR

- Q - 3 (a) Explain the speed control of DC shunt motors along with necessary diagram. [07]

- (b) What is electrical braking of dc motors? Explain the various methods of electrical braking. [07]

SECTION-II

- Q - 4 (a) Give the location (place) and function of following transformer components. [07]

- Buchholz relay
- Breather
- Conservator tank
- L.V. & H.V. winding
- Bushings
- Core of cell type transformer
- Transfer oil

- Q - 5 (a) Write a brief technical note on transformer losses. [07]

- (b) With the schematic diagram explain Sumpner's test. Also, write the importance of this test. [07]

OR

- Q - 5 (a) Explain the transformer equivalent circuit referred to primary side with necessary diagram. [07]

- (b) A 10 KVA, 200/400 V, 50 Hz single phase transformer gave the following test results: OC test (HV winding open) : 200 V 1.3 A 120 W.
SC test (LV winding short-circuited) : 22 V 30 A 200 W.

Find parameters of equivalent circuit as referred to LV winding.

- Q - 6 (a) Briefly explain the open circuit and short circuit test. Which losses can be calculated from this test? How? [07]

- (b) The primary and secondary windings of a 500 KVA, 1-phase transformer have resistances of 0.4 ohm and 0.0015 ohm respectively. The primary and secondary voltages are 6000 V and 400 V respectively and the iron losses is 3.2 kW. Calculate the efficiency on (i) Full load, (ii) Half load, Assuming the pf of the load to be 0.8. [07]

OR

- Q - 6 (a) With neat and clean diagram explain the 3 phase to 2 phase conversion. [07]

- (b) Write a short note on: [07]

- a) On-load tap changer b) Transformer Noise.
